We will show that for every $\delta > 0$, with an appropriate choice of γ, SquareCB (that is, the algorithm that chooses $p^t = \mathsf{IGW}_\gamma(\widehat{f}^t(x^t, \cdot)))$ ensures that with probability at least $1 - \delta$,

$$\mathbf{Reg} \lesssim \sqrt{AT \cdot (\mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + \log(1/\delta))} + \varepsilon \cdot A^{1/2}T.$$

Assume that all functions in $\mathcal{F}$ and rewards take values in $[0, 1]$.

1. Show that for any sequence of estimators $\widehat{f}^1, \ldots, \widehat{f}^t$, by choosing $p^t = \mathsf{IGW}_\gamma(\widehat{f}^t(x^t, \cdot))$, we have that

$$\mathbf{Reg} = \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}[f^\star(x^t, \pi^\star(x^t)) - f^\star(x^t, \pi^t)] \lesssim \frac{AT}{\gamma} + \gamma \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] + \varepsilon T.$$

If we had $f^\star = \bar{f}$, this would follow from Proposition 9, but the difference is that in general ($\bar{f} \neq f^\star$), the expression above measures estimation error with respect to the best-in-class model $\bar{f}$ rather than the true model $f^\star$ (at the cost of an extra εT factor).

2. Show that the following inequality holds for every sequence

$$\sum_{t=1}^{T} (\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2 \leq \mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + 2\sum_{t=1}^{T} (r^t - \bar{f}(x^t, \pi^t))(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t)).$$

3. Using Freedman's inequality (Lemma 35), show that with probability at least $1 - \delta$,

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] \leq 2\sum_{t=1}^{T} (\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2 + O(\log(1/\delta)).$$

4. Using Freedman's inequality once more, show that with probability at least $1 - \delta$,

$$2\sum_{t=1}^{T} (r^t - \bar{f}(x^t, \pi^t))(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t)) \leq \frac{1}{4}\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] + O(\varepsilon^2 T + \log(1/\delta)).$$

Conclude that with probability at least $1 - \delta$,

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] \lesssim \mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + \varepsilon^2 T + \log(1/\delta).$$

5. Combining the previous results, show that for any $\delta > 0$, by choosing $\gamma > 0$ appropriately, we have that with probability at least $1 - \delta$,

$$\mathbf{Reg} \lesssim \sqrt{AT \cdot (\mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + \log(1/\delta))} + \varepsilon \cdot A^{1/2}T.$$

4. STRUCTURED BANDITS

Up to this point, we have focused our attention on bandit problems (with or without contexts) in which the decision space Π is a small, finite set. This section introduces the *structured bandit* problem, which generalizes the basic (non-contextual) multi-armed bandit problem by allowing for large, potentially infinite or continuous decision spaces. The protocol for the setting is as follows.

Structured Bandit Protocol

for $t = 1, \ldots, T$ **do**

Select decision $\pi^t \in \Pi$. // Π is large and potentially continuous.

Observe reward $r^t \in \mathbb{R}$.

This protocol is exactly the same as for multi-armed bandits (Section 2), except that we have removed the restriction that $\Pi = \{1, \ldots, A\}$, and now allow it to be arbitrary. This added generality is natural in many applications:

- In medicine, the treatment may be a continuous variable, such as a dosage. The treatment could even by a high-dimensional vector (such as dosages for many different medications). See Figure 7.
- In pricing applications, a seller might aim to select a continuous price or vector or prices in order to maximize their returns.
- In routing applications, the decision space may be finite, but combinatorially large. For example, the decision might be a path or flow in a graph.

Both contextual bandits and structured bandits generalize the basic multi-armed bandit problem, by incorporating function approximation and generalization, but in different ways:

- The contextual bandit formulation in Section 3 assumes structure in the context space. The aim here was to *generalize across contexts*, but we restricted the decision space to be finite (unstructured).
- In structured bandits, we will focus our attention on the case of *no contexts*, but will assume the decision space is structured, and aim to *generalize across decisions*.

Clearly, both ideas above can be combined, and we will touch on this in Section 4.5.

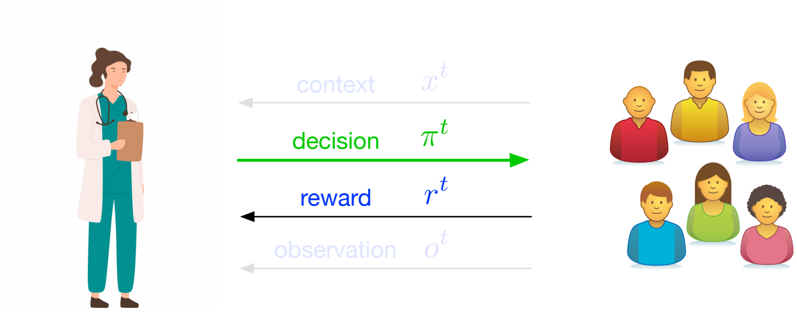

Figure 7: An illustration of the structured bandit problem. A doctor aims to select a continuous, high-dimensional treatment.

Assumptions and regret. To build intuition as to what it means to generalize across decisions, and to give a sense for what sort of guarantees we might hope to prove, let us first give the formal setup for the structured bandit problem. As in preceding sections, we will assume that rewards are stochastic, and generated from a fixed model.

Assumption 4 (Stochastic Rewards): Rewards are generated independently via

$$r^t \sim M^\star(\cdot \mid \pi^t), \tag{4.1}$$

where $M^\star(\cdot \mid \cdot)$ is the underlying *model*.

We define

$$f^\star(\pi) := \mathbb{E}\left[r \mid \pi\right] \tag{4.2}$$

as the mean reward function under $r \sim M^\star(\cdot \mid \pi)$, and measure regret via

$$\mathbf{Reg} := \sum_{t=1}^{T} f^\star(\pi^\star) - \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}[f^\star(\pi^t)]. \tag{4.3}$$

Here, $\pi^\star := \arg\max_{\pi \in \Pi} f^\star(\pi)$ as usual. We will define the history as $\mathcal{H}^t = (\pi^1, r^1), \ldots, (r^t, \pi^t)$.

Function approximation. A first attempt to tackle the structured bandit problem might be to apply algorithms for the multi-armed bandit setting, such as UCB. This would give regret $\widetilde{O}(\sqrt{|\Pi|T})$, which could be vacuous if Π is large relative to T. However, with no further assumptions on the underlying reward function $f^\star$, this is unavoidable. To allow for better regret, we will make assumptions on the structure of $f^\star$ that will allow us to share information across decisions, and to generalize to decisions that we may not have played. This is well-suited for the applications described above, where Π is a continuous set (e.g., $\Pi \subseteq \mathbb{R}^d$), but we expect $f^\star$ to be continuous, or perhaps even linear with respect some well-designed set of features. To make this idea precise, we follow the same approach as in statistical learning and contextual bandits, and assume access to a *well-specified* function class $\mathcal{F}$ that aims to capture our prior knowledge about $f^\star$.

Assumption 5: The decision-maker has access to a class $\mathcal{F} \subset \{f : \Pi \to \mathbb{R}\}$ such that $f^\star \in \mathcal{F}$.

Given such a class, a reasonable goal—particularly in light of the development in Section 1 and Section 3—would be to achieve guarantees that scale with the complexity of supervised learning or estimation with $\mathcal{F}$, e.g. $\log|\mathcal{F}|$ for finite classes; this is what we were able to achieve for contextual bandits, after all. Unfortunately, this is too good to be true, as the following example shows.

Example 4.1 (Necessity of structural assumptions)**.** Let $\Pi = [A]$, and let $\mathcal{F} = \{f_i\}_{i \in [A]}$, where

$$\text{Reg} \gtrsim \tfrac{1}{2} \times A = \Omega(A) \quad f_i(\pi) := \frac{1}{2} + \frac{1}{2}\mathbb{I}\left\{\pi = i\right\}.$$

It is clear that one needs $\mathbf{Reg} \gtrsim A$ for this setting, yet $\log|\mathcal{F}| = \log(A)$, so a regret bound of the form $\mathbf{Reg} \lesssim \sqrt{T \log|\mathcal{F}|}$ is not possible if A is large relative to T. ◁

What this example highlights is that generalizing across decisions is fundamentally different (and, in some sense, more challenging) than generalizing across contexts. In light of this, we will aim for guarantees that scale with $\log|\mathcal{F}|$, but additionally scale with an appropriate notion of *complexity of exploration* for the decision space Π. Such a notion of complexity should reflect how much information is shared across decisions, which depends on the interplay between Π and $\mathcal{F}$.

4.1 Building Intuition: Optimism for Structured Bandits

Our goal is to obtain regret bounds for structured bandits that reflect the intrinsic difficulty of exploring the decision space Π, which should reflect the structure of the function class $\mathcal{F}$ under consideration. To build intuition as to what such guarantees will look like, and how they can be obtained, we first investigate the behavior of the optimism principle and the UCB algorithm when applied to structured bandits. We will see that:

1. UCB attains guarantees that scale with $\log|\mathcal{F}|$, and additionally scale with a notion of complexity called the *eluder dimension*, which is small for simple problems such as bandits with linear rewards.

2. In general, UCB is not optimal, and can have regret that is exponentially large compared to the optimal rate.

4.1.1 UCB for Structured Bandits

We can adapt the UCB algorithm from multi-armed bandits to structured bandit by appealing to least squares and confidence sets, similar to the approach we took for contextual bandits [74]. Assume $\mathcal{F} = \{f : \Pi \to [0,1]\}$ and $r^t \in [0,1]$ almost surely. Let

$$\widehat{f}^t = \arg\min_{f\in\mathcal{F}} \sum_{i=1}^{t-1} (f(\pi^i) - r^i)^2 \tag{4.4}$$

Freedman's inequality

be the empirical minimizer on round t, and with $\beta := 8\log(|\mathcal{F}|/\delta)$, define confidence sets $\mathcal{F}^1 = \mathcal{F}$ and

$$\mathcal{F}^t = \left\{ f \in \mathcal{F} : \sum_{i=1}^{t-1} (f(\pi^i) - r^i)^2 \le \sum_{i=1}^{t-1} (\widehat{f}^t(\pi^i) - r^i)^2 + \beta \right\}. \tag{4.5}$$

Defining $\bar{f}^t(\pi) := \max_{f\in\mathcal{F}^t} f(\pi)$ as the upper confidence bound, the generalized UCB algorithm is given by

$$\pi^t = \arg\max_{\pi\in\Pi} \bar{f}^t(\pi). \tag{4.6}$$

When does the confidence width shrink? Using Proposition 7, one can see the generalized UCB algorithm ensures that $f^\star \in \mathcal{F}^t$ for all t with high probability. Whenever this happens, regret is bounded by the upper confidence width:

$$\mathbf{Reg} \le \sum_{t=1}^{T} \bar{f}^t(\pi^t) - f^\star(\pi^t). \quad \text{(Equation 2.17)} \tag{4.7}$$

This bound holds for all structured bandit problems, with no assumption on the structure of Π and $\mathcal{F}$. Hence, to derive a regret bound, the only question we need to answer is *when will the confidence widths shrink?*

Proposition 5

For the unstructured multi-armed bandit, we need to shrink the width for every arm separately, and the best bound on (4.7) we can hope for is $O(\sqrt{|\Pi|T})$. One might hope that if Π and $\mathcal{F}$ have nice structure, we can do better. In fact, we have already seen one such case: For linear models, where

$$\mathcal{F} = \Big\{\pi \mapsto \langle \theta, \phi(\pi) \rangle \mid \theta \in \Theta \subset \mathsf{B}_2^d(1)\Big\}, \tag{4.8}$$

Proposition 7 shows that we can bound (4.7) by $\sqrt{dT \log|\mathcal{F}|}$. Here, the number of decisions $|\Pi|$ is replaced by the dimension d, which reflects the fact that there are only d truly unique directions to explore before we can start *extrapolating* to new actions. Is there a more general version of this phenomenon when we move beyond linear models?

4.1.2 The Eluder Dimension

The eluder dimension [74] is a complexity measure that aims to capture the extent to which the function class $\mathcal{F}$ facilitates extrapolation (i.e., generalization to unseen decisions), and gives a generic way of bounding the confidence width in (4.7). It is defined for a class $\mathcal{F}$ as follows.

Edim lower ⇒ generalize well to new decision based on previously seen data

Definition 6 (Eluder Dimension): Let $\mathcal{F} \subset (\Pi \to \mathbb{R})$ and $f^\star : \Pi \to \mathbb{R}$ be given, and define $\underline{\mathsf{Edim}}_{f^\star}(\mathcal{F}, \varepsilon)$ as the length of the longest sequence of decisions $\pi^1, \ldots, \pi^d \in \Pi$ such that for all $t \in [d]$, there exists $f^t \in \mathcal{F}$ such that

$$|f^t(\pi^t) - f^\star(\pi^t)| > \varepsilon, \quad \text{and} \quad \sum_{i<t} (f^t(\pi^i) - f^\star(\pi^i))^2 \le \varepsilon^2. \tag{4.9}$$

The eluder dimension is defined as $\mathsf{Edim}_{f^\star}(\mathcal{F}, \varepsilon) = \sup_{\varepsilon' \ge \varepsilon} \underline{\mathsf{Edim}}_{f^\star}(\mathcal{F}, \varepsilon') \vee 1$. We abbreviate $\mathsf{Edim}(\mathcal{F}, \varepsilon) = \max_{f^\star \in \mathcal{F}} \mathsf{Edim}_{f^\star}(\mathcal{F}, \varepsilon)$.

The intuition behind the eluder dimension is simple: It asks, for a worst-case sequence of decisions, how many times we can be "surprised" by a new decision π^t if we can estimate the underlying model $f^\star$ well on all of the preceding points. In particular, if we form confidence sets as in (4.5) with $\beta = \varepsilon^2$, then the number of times the upper confidence width in (4.7) can be larger than ε is at most $\mathsf{Edim}_{f^\star}(\mathcal{F}, \varepsilon)$. We consider the definition $\mathsf{Edim}_{f^\star}(\mathcal{F}, \varepsilon) = \sup_{\varepsilon' \ge \varepsilon} \underline{\mathsf{Edim}}_{f^\star}(\mathcal{F}, \varepsilon') \vee 1$ instead of directly working with $\underline{\mathsf{Edim}}_{f^\star}(\mathcal{F}, \varepsilon)$ to ensure monotonicity with respect to ε, which will be useful in the proofs that follow.

want to ensure

if $\varepsilon_1 < \varepsilon_2 \Rightarrow \mathrm{Edim}(\varepsilon_1) \ge \mathrm{Edim}(\varepsilon_2)$

1) If ε is big, small difference between $f^t(\pi^t)$ and $f^*(\pi^t)$ are ignored. ⇒ easier generalization ⇒ low Edim

2) So we expect $\varepsilon \downarrow \Rightarrow \mathrm{Edim} \downarrow$ however, $\varepsilon \downarrow \Rightarrow$ makes $\Sigma(\ldots)^2 \le \varepsilon^2$ hard to be satisfied which lowers Edim

3) $\varepsilon_1 < \varepsilon_2 \Rightarrow \{\varepsilon' \ge \varepsilon_2\} \subset \{\varepsilon \ge \varepsilon_1\}$

The following result gives a regret bound for UCB for generic structured bandit problems. The regret bound has no dependence on the size of the decision space, and scales only with $\mathsf{Edim}(\mathcal{F}, \varepsilon)$ and $\log|\mathcal{F}|$.

Proposition 12: For a finite set of functions $\mathcal{F} \subset (\Pi \to [0,1])$, using $\beta = 8\log(|\mathcal{F}|/\delta)$, the generalized UCB algorithm guarantees that with probability at least $1-\delta$,

$$\mathbf{Reg} \lesssim \min_{\varepsilon > 0} \Big\{ \sqrt{\mathsf{Edim}(\mathcal{F}, \varepsilon) \cdot T \log(|\mathcal{F}|/\delta)} + \varepsilon T \Big\} \lesssim \sqrt{\mathsf{Edim}(\mathcal{F}, T^{-1/2}) \cdot T \log(|\mathcal{F}|/\delta)}. \tag{4.10}$$

For the case of linear models in (4.8), it is possible to use the elliptic potential lemma (Lemma 11) to show that

$$\mathsf{Edim}(\mathcal{F},\varepsilon) \lesssim d\log(\varepsilon^{-1}).$$

For finite classes, this gives $\mathbf{Reg} \lesssim \sqrt{dT\log(|\mathcal{F}|/\delta)\log(T)}$, which recovers the guarantee in Proposition 7. Another well-known example is that of *generalized linear models*. Here, we fix *link function* $\sigma : [-1,+1] \to \mathbb{R}$ and define

$$\mathcal{F} = \Big\{\pi \mapsto \sigma\big(\langle\theta,\phi(\pi)\rangle\big) \mid \theta \in \Theta \subset \mathsf{B}_2^d(1)\Big\}.$$

This is a more flexible model than linear bandits. A well-known special case is the logistic bandit problem, where $\sigma(z) = 1/(1+e^{-z})$. One can show [74] that for any choice of σ, if there exist $\mu, L > 0$ such that $\mu < \sigma'(z) < L$ for all $z \in [-1,+1]$, then

$$\mathsf{Edim}(\mathcal{F},\varepsilon) \lesssim \frac{L^2}{\mu^2}\cdot d\log(\varepsilon^{-1}). \tag{4.11}$$

This leads to a regret bound that scales with $\frac{L}{\mu}\sqrt{dT\log|\mathcal{F}|}$, generalizing the regret bound for linear bandits.

In general, the eluder dimension can be quite large. Consider the generalized linear model setup above with $\sigma(z) = +\mathsf{relu}(z)$ or $\sigma(z) = -\mathsf{relu}(z)$ (either choice of sign works), where $\mathsf{relu}(z) := \max\{z, 0\}$ is the ReLU function; this can be interpreted as a neural network with a single neuron. Here, we can have $\sigma'(z) = 0$, so (4.11) does not apply, and it turns out [61] that

$$\mathsf{Edim}(\mathcal{F},\varepsilon) \gtrsim e^d \tag{4.12}$$

for constant ε. That is, even for a single ReLU neuron, the eluder dimension is already exponential, which is a bit disappointing. Fortunately, we will show in the sequel that the eluder dimension can be overly pessimistic, and it is possible to do better, but this will require changing the algorithm.

Proof of Proposition 12. Define

$$\bar{\mathcal{F}}^t = \left\{f \in \mathcal{F} \mid \sum_{i<t}(f(\pi^i) - f^\star(\pi^i))^2 \le 4\beta\right\}.$$

By Lemma 10, we have that with probability at least $1-\delta$, for all t:

1. $f^\star \in \mathcal{F}^t$.
2. $\mathcal{F}^t \subseteq \bar{\mathcal{F}}^t$.

Let us condition on this event. As in Lemma 7 , since $f^\star \in \mathcal{F}^t$, we can upper bound

$$\mathbf{Reg} \le \sum_{t=1}^T \bar{f}^t(\pi^t) - f^\star(\pi^t).$$

Now, define

$$w^t(\pi) = \sup_{f\in\bar{\mathcal{F}}^t}\,[f(\pi) - f^\star(\pi)],$$

which is a useful upper bound on the upper confidence width at time t. Since $\mathcal{F}^t \subseteq \overline{\mathcal{F}}^t$, we have

$$\mathbf{Reg} \le \sum_{t=1}^{T} w^t(\pi^t).$$

We now appeal to the following technical lemma concerning the eluder dimension.

Lemma 12 (Russo and Van Roy [74], Lemma 3): Fix a function class $\mathcal{F}$, function $f^\star \in \mathcal{F}$, and parameter $\beta > 0$. For any sequence $\pi^1, \ldots, \pi^T$, if we define

$$w^t(\pi) = \sup_{f \in \mathcal{F}} \left\{ f(\pi) - f^\star(\pi) : \sum_{i<t} (f(\pi^i) - f^\star(\pi^i))^2 \le \beta \right\},$$

then for all $\alpha > 0$,

$$\sum_{t=1}^{T} \mathbb{I}\{w^t(\pi^t) > \alpha\} \le \left(\frac{\beta}{\alpha^2} + 1\right) \cdot \underline{\mathsf{Edim}}_{f^\star}(\mathcal{F}, \alpha).$$

Note that for the special case where $\beta = \alpha^2$, the bound in Lemma 12 immediately follows from the definition of the eluder dimension. The point of this lemma is to show that a similar bound holds for all scales α simultaneously, but with a pre-factor $\frac{\beta}{\alpha^2}$ that grows large when $\alpha^2 \ll \beta$.

To apply this result, fix $\varepsilon > 0$, and bound

$$\sum_{t=1}^{T} w^t(\pi^t) \le \sum_{t=1}^{T} w^t(\pi^t)\mathbb{I}\{w^t(\pi^t) > \varepsilon\} + \varepsilon T. \tag{4.13}$$

Let us order the indices $\{1, \ldots, T\}$ as $\{i_1, \ldots, i_T\}$, so that $w^{i_1}(\pi^{i_1}) \ge w^{i_2}(\pi^{i_2}) \ge \ldots \ge w^{i_\tau}(\pi^{i_\tau})$. Consider any index τ for which $w^{i_\tau}(\pi^{i_\tau}) > \varepsilon$. For any $\alpha > \varepsilon$, if we have $w^{i_\tau}(\pi^{i_\tau}) > \alpha$, then Lemma 12 (since $\alpha \le 1 \le \beta$) implies that

$$\tau \le \sum_{t=1}^{T} \mathbb{I}\{w^t(\pi^t) > \alpha\} \le \left(\frac{4\beta}{\alpha^2} + 1\right) \underline{\mathsf{Edim}}_{f^\star}(\mathcal{F}, \alpha) \le \frac{5\beta}{\alpha^2} \underline{\mathsf{Edim}}_{f^\star}(\mathcal{F}, \alpha). \tag{4.14}$$

Since we have restricted to $\alpha \ge \varepsilon$ and $\alpha \mapsto \mathsf{Edim}_{f^\star}(\mathcal{F}, \alpha)$ is decreasing, rearranging yields

$$w^{i_\tau}(\pi^{i_\tau}) \le \sqrt{\frac{5\beta \mathsf{Edim}(\mathcal{F}, \varepsilon)}{\tau}}.$$

With this, we can bound the main term in (4.13) by

$$\sum_{t=1}^{T} w^t(\pi^t)\mathbb{I}\{w^t(\pi^t) > \varepsilon\} \lesssim \sum_{t=1}^{T} \sqrt{\frac{\beta \mathsf{Edim}(\mathcal{F}, \varepsilon)}{t}} \lesssim \sqrt{\beta \mathsf{Edim}(\mathcal{F}, \varepsilon) T}.$$

Combining this with (4.13) gives $\mathbf{Reg} \lesssim \sqrt{\beta \mathsf{Edim}(\mathcal{F}, \varepsilon) T} + \varepsilon T$. Since $\varepsilon > 0$ was arbitrary, we are free to minimize over it.

□

Proof of Lemma 12. Let us adopt the shorthand $d = \underline{\mathsf{Edim}}_{f^\star}(\mathcal{F}, \alpha)$. We begin with a definition. We say π is α-independent of $\pi^1, \ldots, \pi^t$ if there exists $f \in \mathcal{F}$ such that $|f(\pi) - f^\star(\pi)| > \alpha$ and $\sum_{i=1}^t (f(\pi^i) - f^\star(\pi^i))^2 \leq \alpha^2$. We say π is α-dependent on $\pi^1, \ldots, \pi^t$ if for all $f \in \mathcal{F}$ with $\sum_{i=1}^t (f(\pi^i) - f^\star(\pi^i))^2 \leq \alpha^2$, $|f(\pi) - f^\star(\pi)| \leq \alpha$.

We first claim that for any t, if $w^t(\pi^t) > \alpha$, then π_t is α-dependent on at most β/α^2 disjoint subsequences of $\pi^1, \ldots, \pi^{t-1}$. Indeed, let f be such that $|f(\pi^t) - f^\star(\pi^t)| > \alpha$. If π^t is α-dependent on a particular subsequence $\pi^{i_1}, \ldots, \pi^{i_k}$ but $w^t(\pi^t) > \alpha$, we must have

$$\sum_{j=1}^{k} (f(\pi^{i_j}) - f^\star(\pi^{i_j}))^2 \geq \alpha^2.$$

If there are M such disjoint sequences, we have

$$M\alpha^2 \leq \sum_{i<t} (f(\pi^i) - f^\star(\pi^i))^2 \leq \beta,$$

so $M \leq \frac{\beta}{\alpha^2}$.

Next, we claim that for τ and any sequence $(\pi^1, \ldots, \pi^\tau)$, there is some j such that π^j is α-dependent on at least $\lfloor \tau/d \rfloor$ disjoint subsequences of $\pi^1, \ldots, \pi^{j-1}$. Let $N = \lfloor \tau/d \rfloor$, and let $B_1, \ldots, B_N$ be subsequences of $\pi^1, \ldots, \pi^\tau$. We initialize with $B_i = (\pi^i)$. If π^{N+1} is α-dependent on $B_i = (\pi^i)$ for all $1 \leq i \leq N$ we are done. Otherwise, choose i such that π^{N+1} is α-independent of B_i, and add it to B_i. Repeat this process until we reach j such that either π^j is α-dependent on all B_i or $j = \tau$. In the first case we are done, while in the second case, we have $\sum_{i=1}^N |B_i| \geq \tau \geq dN$. Moreover, $|B_i| \leq d$, since each $\pi^j \in B_i$ is α-independent of its prefix (this follows from the definition of eluder dimension). We conclude that $|B_i| = d$ for all i, so in this case π^τ is α-dependent on all B_i.

Finally, let $(\pi^{t_1}, \ldots, \pi^{t_\tau})$ be the subsequence $\pi^1, \ldots, \pi^T$ consisting of all elements for which $w^{i_i}(\pi^{t_i}) > \alpha$. Each element of the sequence is dependent on at most β/α^2 disjoint subsequences of $(\pi^{t_1}, \ldots, \pi^{t_\tau})$, and by the argument above, one element is dependent on at least $\lfloor \tau/d \rfloor$ disjoint subsequences, so we must have $\lfloor \tau/d \rfloor \leq \beta/\alpha^2$, and which implies that $\tau \leq (\beta/\alpha^2 + 1)d$. □

4.1.3 Suboptimality of Optimism

The following example shows a function class $\mathcal{F}$ for which the regret experienced by UCB is exponentially large compared to the regret obtained by a simple alternative algorithm. This shows that while the algorithm is useful for some special cases, it does not provide a general principle that attains optimal regret for any structured bandit problem.

Example 4.2 (Cheating Code [9, 50])**.** Let $A \in \mathbb{N}$ be a power of 2 and consider the following function class $\mathcal{F}$.

- The decision space is $\Pi = [A] \cup \mathcal{C}$, where $\mathcal{C} = \{c_1, \ldots, c_{\log_2(A)}\}$ is a set of "cheating" actions.
- For all actions $\pi \in [A]$, $f(\pi) \in [0, 1]$ for all $f \in \mathcal{F}$, but we otherwise make no assumption on the reward.
- For each $f \in \mathcal{F}$, rewards for actions in $\mathcal{C}$ take the following form. Let $\pi_f \in [A]$ denote the action in $[A]$ with highest reward. Let $b(f) = (b_1(f), \ldots, b_{\log_2(A)}(f)) \in \{0, 1\}^{\log_2(A)}$

be a binary encoding for the index of $\pi_f \in [A]$ (e.g., if $\pi_f = 1$, $b(f) = (0, 0, \ldots, 0)$, if $\pi_f = 2$, $b(f) = (0, 0, \ldots, 0, 1)$, and so on). For each action $c_i \in \mathcal{C}$, we set

$$f(c_i) = -b_i(f).$$

The idea here is that if we ignore the actions $\mathcal{C}$, this looks like a standard multi-armed bandit problem, and the optimal regret is $\Theta(\sqrt{AT})$. However, we can use the actions in $\mathcal{C}$ to "cheat" and get an exponential improvement in sample complexity. The argument is as follows.

Suppose for simplicity that rewards are Gaussian with $r \sim \mathcal{N}(f^\star(\pi), 1)$ under π. For each cheating action $c_i \in \mathcal{C}$, since $f^\star(c_i) = -b_i(f^\star) \in \{0, -1\}$, we can determine whether the value is $b_i(f^\star) = 0$ or $b_i(f^\star) = 1$ with high probability using $\widetilde{O}(1)$ action pulls. If we do this for each $c_i \in \mathcal{C}$, which will incur $\widetilde{O}(\log(A))$ regret (there are $\log(A)$ such actions and each one leads to constant regret), we can infer the binary encoding $b(f^\star) = b_1(f^\star), \ldots, b_{\log_2(A)}(f^\star)$ for the optimal action $\pi_{f^\star}$ with high probability. At this point, we can simply stop exploring, and commit to playing $\pi_{f^\star}$ for the remaining rounds, which will incur no more regret. If one is careful with the details, this gives that with probability at least $1 - \delta$,

$$\mathbf{Reg} \lesssim \log^2(A/\delta).$$

In other words, by exploiting the cheating actions, our regret has gone from linear to *logarithmic* in A (we have also improved the dependence on T, which is a secondary bonus).

Now, let us consider the behavior of the generalized UCB algorithm. Unfortunately, since all actions $c_i \in \mathcal{C}$ have $f(c_i) \leq 0$ for all $f \in \mathcal{F}$, we have $\bar{f}^t(c_i) \leq 0$. As a result, the generalized UCB algorithm will only ever pull actions in $[A]$, ignoring the cheating actions and effectively turning this into a vanilla multi-armed bandit problem, which means that

$$\mathbf{Reg} \gtrsim \sqrt{AT}.$$

◁

This example shows that UCB can behave suboptimally in the presence of decisions that reveal useful information but do not necessarily lead to high reward. Since the "cheating" actions are guaranteed to have low reward, UCB avoids them even though they are very informative. We conclude that:

1. Obtaining optimal sample complexity for structured bandits requires algorithms that more deliberately balance the tradeoff between optimizing reward and acquiring information.

2. In general, the optimal strategy for picking decisions can be very different depending on the choice of the class $\mathcal{F}$. This contrasts the contextual bandit setting, where we saw that the Inverse Gap Weighting algorithm attained optimal sample complexity for any choice of class $\mathcal{F}$, and all that needed to change was how to perform estimation.

Remark 13 (Suboptimality of posterior sampling): Recall the Bayesian bandit setting in Section 2.4, where we showed that the posterior sampling algorithm attains regret $\widetilde{O}(\sqrt{AT})$ when $\Pi = \{1, \ldots, A\}$. Posterior sampling is a general-purpose algorithm, and can be applied to directly to arbitrary structured bandit problems (as long as a prior is available). However, similar to UCB, the cheating code construction in

Example 4.2 implies that posterior sampling is not optimal in general. Indeed, posterior sampling will never select the cheating arms in $\mathcal{C}$, as these have sub-optimal reward for all models in $\mathcal{F}$. As a result, the Bayesian regret of the algorithm will scale with $\mathbf{Reg} \gtrsim \sqrt{AT}$ for a worst-case prior.

4.2 The Decision-Estimation Coefficient

The discussion in the prequel highlights two challenges in designing algorithms and understanding sample complexity for structured bandits: 1) the optimal regret (in a sense, the complexity of exploration) can depend on the class $\mathcal{F}$ in a subtle, sometimes surprising fashion, and 2) the algorithms required to achieve optimal regret can heavily depend on the choice of $\mathcal{F}$. In light of these challenges, it is natural to ask whether it is possible to have any sort of unified understanding of the optimal regret. We will now show that the answer is *yes*, and this will be achieved by a single, general-purpose principle for algorithm design.

The algorithm we will present in this section reduces the problem of decision making to that of supervised online learning/estimation, in a similar fashion to the SquareCB method for contextual bandits in Section 3. To apply this method, we require the following oracle for supervised estimation.

Definition 7 (Online Regression Oracle): At each time $t \in [T]$, an *online regression oracle* returns, given

$$(\pi^1, r^1), \ldots, (\pi^{t-1}, r^{t-1})$$

with $\mathbb{E}[r^i|\pi^i] = f^\star(\pi^i)$ and $\pi^i \sim p^i$, a function $\widehat{f}^t : \Pi \to \mathbb{R}$ such that

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}(\widehat{f}^t(\pi^t) - f^\star(\pi^t))^2 \le \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)$$

with probability at least $1-\delta$. Here, $p^i(\cdot|\mathcal{H}^{i-1})$ is the randomization distribution for the decision-maker.

Recall, following the discussion in Section 3, that the averaged exponential weights algorithm achieves is an online regression oracle with $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, t, \delta) \lesssim \log(|\mathcal{F}|/\delta)$.

The following algorithm, which we call Estimation-to-Decisions or E2D [40, 43], is a general-purpose meta-algorithm for structured bandits.

Estimation-to-Decisions (E2D) for Structured Bandits
Input: Exploration parameter $\gamma > 0$.
for $t = 1, \ldots, T$ **do**
Obtain $\widehat{f}^t$ from online regression oracle with $(\pi^1, r^1), \ldots, (\pi^{t-1}, r^{t-1})$.
Compute

$$p^t = \arg\min_{p\in\Delta(\Pi)} \max_{f\in\mathcal{F}} \mathbb{E}_{\pi\sim p}\left[f(\pi_f) - f(\pi) - \gamma\cdot(f(\pi) - \widehat{f}^t(\pi))^2\right].$$

Select action $\pi^t \sim p^t$.

At each timestep t, the algorithm calls invokes an online regression oracle to obtain an estimator $\widehat{f}^t$ using the data $\mathcal{H}^{t-1} = (\pi^1, r^1, \ldots, \pi^{t-1}, r^{t-1})$ observed so far. The algorithm then finds a distribution p^t by solving a min-max optimization problem involving the estimator $\widehat{f}^t$ and the class $\mathcal{F}$, then samples the decision π^t from this distribution.

The minimax problem in E2D is derived from a complexity measure (or, structural parameter) for $\mathcal{F}$ called the *Decision-Estimation Coefficient* [40, 43], whose value is given by

$$\mathsf{dec}_\gamma(\mathcal{F}, \widehat{f}) = \min_{p \in \Delta(\Pi)} \max_{f \in \mathcal{F}} \mathbb{E}_{\pi \sim p}\Bigg[\underbrace{f(\pi_f) - f(\pi)}_{\text{regret of decision}} - \gamma \cdot \underbrace{(f(\pi) - \widehat{f}(\pi))^2}_{\text{information gain for obs.}}\Bigg]. \tag{4.15}$$

The Decision-Estimation Coefficient can be thought of as the value of a game in which the learner (represented by the min player) aims to find a distribution over decisions such that for a worst-case problem instance (represented by the max player), the *regret* of their decision is controlled by a notion of *information gain* (or, estimation error) relative to a reference model $\widehat{f}$. Conceptually, $\widehat{f}$ should be thought of as a guess for the true model, and the learner (the min player) aims to—in the face of an unknown environment (the max player)—optimally balance the regret of their decision with the amount information they acquire. With enough information, the learner can confirm or rule out their guess $\widehat{f}$, and scale parameter γ controls how much regret they are willing to incur to do this. In general, the larger the value of $\mathsf{dec}_\gamma(\mathcal{F}, \widehat{f})$, the more difficult it is to explore.

To state a regret bound for E2D, we define

$$\mathsf{dec}_\gamma(\mathcal{F}) = \sup_{\widehat{f} \in \mathrm{co}(\mathcal{F})} \mathsf{dec}_\gamma(\mathcal{F}, \widehat{f}). \tag{4.16}$$

Here, $\mathrm{co}(\mathcal{F})$ denotes the set of all convex combinations of elements in $\mathcal{F}$. The reason we consider the set $\mathrm{co}(\mathcal{F})$ is that in general, online estimation algorithms such as exponential weights will produce improper predictions with $\widehat{f} \in \mathrm{co}(\mathcal{F})$. In fact, it turns out (see Proposition 24) that even if we allow $\widehat{f}$ to be unconstrained above, the maximizer always lies in $\mathrm{co}(\mathcal{F})$ without loss of generality.

The main result for this section shows that the regret for E2D is controlled by the value of the DEC and the estimation error $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)$ for the online regression oracle.

Proposition 13 (Foster et al. [40]): The E2D algorithm with exploration parameter $\gamma > 0$ guarantees that with probability at least $1 - \delta$,

$$\mathbf{Reg} \leq \mathsf{dec}_\gamma(\mathcal{F}) \cdot T + \gamma \cdot \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta). \tag{4.17}$$

We can optimize over the parameter γ in the result above, which yields

$$\begin{aligned}\mathbf{Reg} &\leq \inf_{\gamma > 0}\left\{\mathsf{dec}_\gamma(\mathcal{F}) \cdot T + \gamma \cdot \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)\right\} \\ &\leq 2 \cdot \inf_{\gamma > 0} \max\left\{\mathsf{dec}_\gamma(\mathcal{F}) \cdot T, \gamma \cdot \mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta)\right\}. \qquad (x+y \leq 2\max\{x,y\})\end{aligned}$$

For finite classes, we can use the exponential weights method to obtain $\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F}, T, \delta) \lesssim \log(|\mathcal{F}|/\delta)$, and this bound specializes to

$$\mathbf{Reg} \lesssim \inf_{\gamma > 0} \max\left\{\mathsf{dec}_\gamma(\mathcal{F}) \cdot T, \gamma \cdot \log(|\mathcal{F}|/\delta)\right\}. \tag{4.18}$$

As desired, this gives a bound on regret that scales only with:

1. the complexity $\log|\mathcal{F}|$ for estimation.

2. the complexity of exploration in the decision space, which is captured by $\mathsf{dec}_\gamma(\mathcal{F})$.

Before interpreting the result further, we give the proof, which is a nearly immediate consequence of the definition of the DEC, and bears strong similarity to the proof of the regret bound for SquareCB (Proposition 10), minus contexts.

Proof of Proposition 13. We write

$$\begin{aligned}\mathbf{Reg} &= \sum_{t=1}^{T} \mathbb{E}_{\pi^t\sim p^t}[f^\star(\pi^\star) - f^\star(\pi^t)] \\ &= \sum_{t=1}^{T} \mathbb{E}_{\pi^t\sim p^t}[f^\star(\pi^\star) - f^\star(\pi^t)] - \gamma\cdot\mathbb{E}_{\pi^t\sim p^t}\Big[(f^\star(\pi^t) - \widehat{f}^t(\pi^t))^2\Big] + \gamma\cdot\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F},T,\delta).\end{aligned}$$

For each t, since $f^\star\in\mathcal{F}$, we have

$$\begin{aligned}&\mathbb{E}_{\pi^t\sim p^t}[f^\star(\pi^\star) - f^\star(\pi^t)] - \gamma\cdot\mathbb{E}_{\pi^t\sim p^t}\Big[(f^\star(\pi^t) - \widehat{f}^t(\pi^t))^2\Big] \\ &\leq \sup_{f\in\mathcal{F}}\Big\{\mathbb{E}_{\pi^t\sim p^t}[f(\pi_f) - f(\pi^t)] - \gamma\cdot\mathbb{E}_{\pi^t\sim p^t}\Big[(f(\pi^t) - \widehat{f}^t(\pi^t))^2\Big]\Big\} \\ &= \inf_{p\in\Delta(\Pi)}\sup_{f\in\mathcal{F}}\mathbb{E}_{\pi\sim p}\Big[f(\pi_f) - f(\pi) - \gamma\cdot(f(\pi^t) - \widehat{f}^t(\pi^t))^2\Big] \\ &= \mathsf{dec}_\gamma(\mathcal{F},\widehat{f}^t),\end{aligned} \tag{4.19}$$

where the first equality above uses that p^t is chosen as the minimizer for $\mathsf{dec}_\gamma(\mathcal{F},\widehat{f}^t)$. Summing across rounds, we conclude that

$$\mathbf{Reg} \leq \sup_{\widehat{f}}\mathsf{dec}_\gamma(\mathcal{F},\widehat{f})\cdot T + \gamma\cdot\mathbf{Est}_{\mathsf{Sq}}(\mathcal{F},T,\delta).$$

□

When designing algorithms for structured bandits, a common challenge is that the connection between decision making (where the learner's decisions influence what feedback is collected) and estimation (where data is collected passively) may not seem apparent a-priori. The power of the Decision-Estimation Coefficient is that it—*by definition*—provides a bridge, which the proof of Proposition 13 highlights. One can select decisions by building an estimate for the model using all of the observations collected so far, then sampling from the distribution p that solves (4.15) with the estimated reward function $\widehat{f}$ plugged in. Boundedness of the DEC implies that at every round, any learner using this strategy either enjoys small regret or acquires information, with their total regret controlled by the cumulative online estimation error.

Example: Multi-Armed Bandit. Of course, the perspective above is only useful if the DEC is indeed bounded, which itself is not immediately apparent. In Section 6, we will show that boundedness of the DEC is not just sufficient, but in fact *necessary* for low regret in a fairly strong quantitative sense. For now, we will build intuition about the DEC through examples. We begin with the multi-armed bandit, where $\Pi = [A]$ and $\mathcal{F} = \mathbb{R}^A$. Our first result shows that $\mathsf{dec}_\gamma(\mathcal{F}) \leq \frac{A}{\gamma}$, and that this is achieved with the Inverse Gap Weighting method introduced in Section 3.

Proposition 14 (IGW minimizes the DEC): For the Multi-Armed Bandit setting, where $\Pi = [A]$ and $\mathcal{F} = \mathbb{R}^A$, the Inverse Gap Weighting distribution $p = \mathsf{IGW}_{4\gamma}(\widehat{f})$ in (3.36) is the *exact* minimizer for $\mathsf{dec}_\gamma(\mathcal{F}, \widehat{f})$, and certifies that $\mathsf{dec}_\gamma(\mathcal{F}, \widehat{f}) = \frac{A-1}{4\gamma}$.

By rewriting Proposition 9, it is straightforward to deduce that the DEC is bounded by $\frac{A}{\gamma}$, but Proposition 14 shows that IGW is actually the *best possible distribution* for this minimax problem. In this sense, the SquareCB algorithm can be seen as a (contextual) special case of the Estimation-to-Decisions principle. Note that to attain the exact optimal value (instead of a bound that is optimal up to constants), we use $\mathsf{IGW}_{4\gamma}$ as opposed IGW_γ as in Proposition 9; the reason why this choice for γ is optimal is related to the fact that the inequality $xy \leq x^2 + \frac{1}{4}y^2$ is tight in general.

Proof of Proposition 14. We rewrite the minimax problem as

$$
\begin{aligned}
&\min_{p\in\Delta([A])} \max_{f\in\mathbb{R}^A} \mathbb{E}_{\pi\sim p}\Big[f(\pi_f) - f(\pi) - \gamma(f(\pi) - \widehat{f}(\pi))^2\Big] \\
&= \min_{p\in\Delta([A])} \max_{f\in\mathbb{R}^A} \max_{\pi^\star\in[A]} \mathbb{E}_{\pi\sim p}\Big[f(\pi^\star) - f(\pi) - \gamma(f(\pi) - \widehat{f}(\pi))^2\Big] \\
&= \min_{p\in\Delta([A])} \max_{\pi^\star\in[A]} \max_{f\in\mathbb{R}^A} \mathbb{E}_{\pi\sim p}\Big[f(\pi^\star) - f(\pi) - \gamma(f(\pi) - \widehat{f}(\pi))^2\Big].
\end{aligned}
$$

For any fixed p and $\pi^\star$, first-order conditions for optimality imply that the choice for f that maximizes this expression is

$$
f(\pi) = \widehat{f}(\pi) - \frac{1}{2\gamma} + \frac{1}{2\gamma p(\pi^\star)}\mathbb{I}\{\pi = \pi^\star\}.
$$

$\left\{ \begin{array}{l} \pi \neq \pi^* \\ \pi = \pi^* \end{array} \right.$ derived by $\frac{d}{df(\pi)}$ on expectation

This choice gives

$$
\mathbb{E}_{\pi\sim p}[f(\pi^\star) - f(\pi)] = \mathbb{E}_{\pi\sim p}\Big[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\Big] + \frac{1 - p(\pi^\star)}{2\gamma p(\pi^\star)}
$$

and

$$
\gamma\,\mathbb{E}_{\pi\sim p}\Big[(f(\pi) - \widehat{f}(\pi))^2\Big] = \frac{1 - p(\pi^\star)}{4\gamma} + \frac{(1 - p(\pi^\star))^2}{4\gamma p(\pi^\star)} = \frac{1}{4\gamma p(\pi^\star)} - \frac{1}{4\gamma}.
$$

Plugging in and simplifying, we compute that the original minimax game is equivalent to

$$
\min_{p\in\Delta([A])} \max_{\pi^\star\in[A]} \left\{ \mathbb{E}_{\pi\sim p}\Big[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\Big] + \frac{1}{4\gamma p(\pi^\star)} \right\} - \frac{1}{4\gamma}. \tag{4.20}
$$

Finishing the proof: Ad-hoc approach. Observe that for any $p \in \Delta(\Pi)$, we have

$$
\max_{\pi^\star\in[A]} \left\{ \mathbb{E}_{\pi\sim p}\Big[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\Big] + \frac{1}{4\gamma p(\pi^\star)} \right\} \geq \mathbb{E}_{\pi^\star\sim p}\left[\mathbb{E}_{\pi\sim p}\Big[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\Big] + \frac{1}{4\gamma p(\pi^\star)}\right] = \frac{A}{4\gamma},
$$

↳ 0

so no p can attain value better than $\frac{A}{4\gamma}$. If we can show that IGW achieves this value, we are done.

Observe that by setting $p = \mathsf{IGW}_{4\gamma}(\widehat{f})$, we have that for all $\pi^\star$,

$$\mathbb{E}_{\pi\sim p}\left[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\right] + \frac{1}{4\gamma p(\pi^\star)} = \mathbb{E}_{\pi\sim p}\left[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\right] + \frac{\lambda}{4\gamma} + \widehat{f}(\widehat{\pi}) - \widehat{f}(\pi^\star) \tag{4.21}$$
$$= \mathbb{E}_{\pi\sim p}\left[\widehat{f}(\widehat{\pi}) - \widehat{f}(\pi)\right] + \frac{\lambda}{4\gamma}.$$

Note that the value on the right-hand side is independent of $\pi^\star$. That is, the inverse gap weighting distribution is an equalizing strategy. This means that for this choice of p, we have

minimax optimal

$$\max_{\pi^\star\in[A]}\left\{\mathbb{E}_{\pi\sim p}\left[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\right] + \frac{1}{4\gamma p(\pi^\star)}\right\} = \min_{\pi^\star\in[A]}\left\{\mathbb{E}_{\pi\sim p}\left[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\right] + \frac{1}{4\gamma p(\pi^\star)}\right\}$$
$$= \mathbb{E}_{\pi^\star\sim p}\left\{\mathbb{E}_{\pi\sim p}\left[\widehat{f}(\pi^\star) - \widehat{f}(\pi)\right] + \frac{1}{4\gamma p(\pi^\star)}\right\} = \frac{A}{4\gamma}.$$

Hence, $p = \mathsf{IGW}_{4\gamma}(\widehat{f})$ achieves the optimal value.

Finishing the proof: Principled approach. We begin by relaxing to $p \in \mathbb{R}^A_+$. Define

$$g_{\pi^\star}(p) = \widehat{f}(\pi^\star) + \frac{1}{4\gamma p(\pi^\star)}.$$

Let $\nu \in \mathbb{R}$ be a Lagrange multiplier and $p \in \mathbb{R}^A_+$, and consider the Lagrangian

$$\mathcal{L}(p,\nu) = g_{\pi^\star}(p) - \sum_\pi p(\pi)\widehat{f}(\pi) + \nu\left(\sum_\pi p(\pi) - 1\right).$$

By the KKT conditions, if we wish to show that $p \in \Delta(\Pi)$ is optimal for the objective in (4.20), it suffices to find ν such that[12]

$$\mathbf{0} \in \partial_p \mathcal{L}(p,\nu),$$

where ∂_p denotes the subgradient with respect to p. Recall that for a convex function $h(x) = \max_y g(x,y)$, we have $\partial_x h(x) = \mathrm{co}(\{\nabla g(x,y) \mid g(x,y) = \max_{y'} g(x,y')\})$. As a result,

$$\partial_p \mathcal{L}(p,\nu) = \nu\mathbf{1} - \widehat{f} + \mathrm{co}(\{\nabla_p g_{\pi^\star}(p) \mid g_{\pi^\star}(p) = \max_{\pi'} g_{\pi'}(p)\}).$$

Now, let $p = \mathsf{IGW}_{4\gamma}(\widehat{f})$. We will argue that $\mathbf{0} \in \partial_p\mathcal{L}(p,\nu)$ for an appropriate choice of ν. By (4.21), we know that $g_\pi(p) = g_{\pi'}(p)$ for all π, π' (p is equalizing), so the expression above simplifies to

$$\partial_p \mathcal{L}(p,\nu) = \nu\mathbf{1} - \widehat{f} + \mathrm{co}(\{\nabla_p g_{\pi^\star}(p)\}_{\pi^\star\in\Pi}). \tag{4.22}$$

Noting that $\nabla_p g_{\pi^\star}(p) = -\frac{1}{4\gamma p^2(\pi^\star)} e_{\pi^\star}$, we compute

$$\boldsymbol{\delta} := \sum_\pi p(\pi) g_\pi(p) = \left\{-\frac{1}{4\gamma p(\pi)}\right\}_{\pi\in\Pi} = \left\{-\frac{\lambda}{4\gamma} - \widehat{f}(\widehat{\pi}) + \widehat{f}(\pi)\right\}_{\pi\in\Pi},$$

which has $\boldsymbol{\delta} \in \mathrm{co}(\{\nabla_p g_{\pi^\star}(p)\}_{\pi^\star\in\Pi})$. By choosing $\nu = \frac{\lambda}{4\gamma} + \widehat{f}(\widehat{\pi})$, we have

$$\nu\mathbf{1} - \widehat{f} + \boldsymbol{\delta} = \mathbf{0},$$

so (4.22) is satisfied.

□

[12] If $p \in \Delta(\Pi)$, the KKT condition that $\frac{d}{d\nu}\mathcal{L}(p,\nu) = 0$ is already satisfied.